

Summaiya Bibi

Machine Learning Engineer

summaiya.ai@gmail.com | +92 346 6577540 | Islamabad, Pakistan | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

Results-driven and detail-oriented Machine Learning Engineer with proven proficiency in Python, PyTorch, TensorFlow, Scikit-Learn, NLP, Large Language Models. AI/ML Engineer with 3+ years of experience developing predictive machine learning models, natural language processing (NLP) pipelines, and generative AI systems using Python, PyTorch, and HuggingFace. Proven track record of improving inference speeds by 40% and deploying scalable models with Docker and FastAPI. Recognized for delivering scalable solutions, optimizing system performance, and driving measurable business impact.

TECHNICAL SKILLS

Python, PyTorch, TensorFlow, Scikit-Learn, NLP, Large Language Models, LangChain, RAG, FastAPI, Docker, AWS, SQL, MLOps, HuggingFace, Pandas, NumPy, Git, Streamlit

EXPERIENCE

Machine Learning Engineer — Cognitive AI Solutions

Islamabad, Pakistan | 2024 – Present

- Engineered low-latency RESTful APIs using FastAPI and Docker, orchestrating model deployments with 99.8% uptime on AWS EC2.
- Collaborated with cross-functional engineering teams to automate CI/CD pipeline for automated model evaluation and data drift monitoring.
- Architected and deployed an end-to-end Retrieval-Augmented Generation (RAG) system using LangChain, FAISS, and Llama 3, boosting internal knowledge retrieval accuracy by 42%.
- Fine-tuned Transformer models (BERT, RoBERTa) for multi-class intent classification, achieving an F1-score of 0.94 across 50,000+ customer records.

Junior AI Developer — DataTech Analytics

Rawalpindi, Pakistan | 2022 – 2024

- Developed and evaluated predictive machine learning models using Scikit-Learn and XGBoost for customer churn prediction, reducing churn by 18%.
- Built interactive real-time data visualization dashboards in Streamlit to present model inference insights to executive stakeholders.
- Performed exploratory data analysis (EDA) and automated feature engineering pipelines across tabular datasets of over 2M records using Pandas and NumPy.

PROJECTS

Enterprise Document QA System (RAG Pipeline)

Engineered an AI-powered conversational document intelligence platform indexing thousands of technical PDFs with semantic search and citation verification.

Technologies: Python, LangChain, PyTorch, ChromaDB, FastAPI, Docker

Automated Medical Text Entity Extraction

Trained custom Named Entity Recognition (NER) models using SpaCy and Transformers to extract critical diagnostic entities from clinical notes with 91% precision.

Technologies: Python, HuggingFace, SpaCy, Scikit-Learn, Streamlit

EDUCATION

- BS Computer Science — Virtual University of Pakistan (2021 – 2025)

CERTIFICATIONS

- Deep Learning Specialization (DeepLearning.AI)
- AWS Certified Machine Learning Specialty
- Generative AI & LLMs in Production